

Sofiane CHALAL

Paris (France) | firstname.lastname@centralesupelec.fr | +33781903284

Bio

My research covers an interdisciplinary topic between probability theory, quantum physics and control theory. My work deals with observed quantum particle systems and their mean-field graph limits.

Education

CentraleSupélec | Paris-Saclay *PHD* : Applied mathematics 2026
Dauphine | PSL *Master* : Applied and Theoretical Mathematics - Probability 2021

Research

Journal articles & preprints

Quantitative propagation of chaos for Lindblad dynamics 2026
with : Nina Hadis Amini | [Journal of Statistical Physics](#)

On the mean-field Belavkin filtering equation 2023
with : Nina Hadis Amini, Gaoyue Guo | [IEEE Control Systems Letters](#)

On brownian motion on the Fubini extension space 2025
with : Hamed Amini, Nina Hadis Amini, Gaoyue Guo | [Submitted to Stochastic Processes and their Applications](#)

Graphon quantum filtering system 2025
with : Hamed Amini, Nina Hadis Amini, Gaoyue Guo | [Submitted to Annales Henri Poincaré - A](#)

Semi-Classical limits for Belavkin filtering equation 2026
with : Nina Hadis Amini, | In preparation

Mean-field limit for Hudson-Parthasarathy equation 2026
with : Nina Hadis Amini | In preparation

Mean-field Monte-Carlo approach for N -body open quantum system 2026
with : Nina Hadis Amini | In preparation

On modeling quantum differential and mean-field games 2026
with : Hamed Amini, Nina Hadis Amini, Mathieu Laurière | In preparation

Conference proceedings

Dynamic programming principle and stabilization for mean-field quantum filtering systems 2026
with : Hamed Amini, Nina Hadis Amini, Mathieu Laurière | [IEEE qCCL Aalborg](#)

Quantum approach for electrical system maintenance scheduling problem 2026
with : Titouan Carrette, Fikri Hafid, Nouredine Henka, Sami Tazi, Djahou Tognon, Qinyan Zhou | [ESAIM: Proceedings & Surveys](#)

Unravelling mean-field Lindblad equation 2025
with : Nina Hadis Amini | [IEEE qCCL Hong-Kong](#)

Observed quantum particles system with graphon interaction 2024
with : Hamed Amini, Nina Hadis Amini, Gaoyue Guo | [63rd IEEE CDC](#)

Teaching and Missions

TD : Probabilités numériques Third year, ESILV, 32h	Semester 2 - 2025/2026
Project supervision : Quantum annealing for portfolio optimization Third year, CentraleSupélec, 10h	Semester 1 - 2025/2026
TD : Statistiques & Outils mathématiques First year, IUT Université Paris-Saclay, 32h	Semester 2 - 2024/2025
Project supervision : QUBO and Quantum Annealing Third year, CentraleSupélec, 10h	Semester 1 - 2024/2025
TD : Equations aux dérivées partielles First year, CentraleSupélec, 12h	Semester 1 - 2023/2024
TD : Integration, Convergence et Probabilités First year, CentraleSupélec, 12h	Semester 1 - 2023/2024
Research mission : Quantum annealing and optimization RTE France, 64h	Semester 2 - 2022/2023

Visit

ICTS - Bengaluru, INDIA Quantum Trajectories.	January - February 2025
IMSI - The University of Chicago, USA Statistical Methods and Mathematical Analysis for Quantum Information Science.	November - December 2024

Invited talks

2nd IEEE Conference on Quantum Control, Computing and Learning - Aalborg, DENMARK Dynamic programming principle and stabilization for mean-field quantum filtering systems	July 2026
Workshop RTE Chair at CentraleSupélec - Saclay, FRANCE Quantum approach for electrical system maintenance scheduling problem	June 2026
2nd Dolomites WS, Mean-field systems in finance, neurosciences and AI - Folgarida, ITALY Observed quantum particles system with mean-field graphon interaction	January 2026
IEEE Conference on Quantum Control, Computing and Learning - Hong-Kong Unravelling mean-field Lindblad equation	June 2025
63rd IEEE Conference on Decision and Control - Milan, ITALY Observed quantum particles system with graphon interaction	December 2024
26th International Symposium on Mathematical Theory of Networks and Systems - Cambridge, UK Control and stabilization of mean-field Belavkin equation	August 2024
Séminaire d'Automatique du Plateau de Saclay - Saclay, FRANCE Quantum mean-field filtering and control	February 2024
62nd IEEE Conference on Decision and Control - Singapore On the mean-field filtering Belavkin equation	December 2023
GE2MI conference - Levico, ITALY Mean-field Belavkin equation and quantum feedback control	October 2023
Workshop RTE Chair at CentraleSupélec - Saclay, FRANCE Quantum annealing approach to electrical grids problems	May 2023

Prizes

Best student paper IEEE Conference on Quantum Control, Computing and Learning - Aalborg, 2026.

Skills & Interests

IT : Python (Qutip, Qiskit)

Mathematics : Stochastic Calculus, Stochastic filtering, Monte-Carlo Methods

Interests : I am interested in and enjoy discussing the foundations of quantum mechanics, in particular the measurement problem and the different approaches proposed to address it. For now, I am more convinced by the *eventum mechanics* approach, while remaining open to other viewpoints.